

# Auditing Search-Budget Portability Across Graph-ANNS Rebuilds: [Experiments & Analysis]

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## Abstract

Graph indexes are rebuilt to maintain search quality, yet the budget policy used to serve queries is often treated as reusable state. We audit that assumption through controlled rebuilds, a 5% data-refresh case, and a fresh-query recovery test. On unchanged SIFT-100K and Arxiv-Nomic-100K collections, transferring retrospective first-passing budgets across hnsplib and Faiss builds adds 17.17–23.60 percentage points of query failure relative to target-specific references. Index-Conditioned Budget Audit (ICBA) separates response evidence from executable-policy risk, independent certification, and acquisition cost. After refresh, an unchanged recurring-query policy rises from 3.77% to 6.69% failure on SIFT and from 3.19% to 7.24% on Arxiv-Nomic. Target-calibrated pooling (TCP) retains jointly controlled mean-distance reductions of 44.39% and 29.86%, with cold-history break-even near 396,000 and 457,000 executions. In a separately preregistered fresh-query test, all 24 Faiss source decisions per dataset are independently certified and evaluated over 552 directed target builds. Their one-rung policies save 8.27% and 29.09% mean distance work at held-out empirical risks of 0.18% and 0.67%; endpoint clipping leaves no economic gain on either hnsplib dataset. Thus portability, qualification, and economic recovery are distinct target-dependent decisions.

## Keywords

approximate nearest-neighbor search, graph indexes, index rebuilding, risk auditing, budget transfer

## 1 Introduction

A graph index and its search budget jointly determine the quality and cost of approximate nearest-neighbor search (ANNS). Hierarchical Navigable Small World (HNSW) graphs and Vamana make this tradeoff central to efficient vector retrieval [12, 14]. Index maintenance changes one side of this contract. Even with the same vectors and query, construction order and randomization can change the traversed graph; after data refresh, both the graph and nearest-neighbor ground truth can change. A budget retained from an old index therefore needs its own portability assessment.

The operational question is not simply whether rebuilding changes average recall. A service may require a high fraction of queries to reach a specified recall, while an efficient policy fails that requirement on a subset of requests. We measure the probability of  $\text{Recall}@10 < 0.95$ : because recall is discrete, one missing neighbor is a failure. A 5% query-risk limit asks that at least 95% of queries recover all ten neighbors. It differs from mean recall of 0.95 and from expected missed-neighbor fraction.

Our study separates three forms of evidence. First, on unchanged collections, we transfer a query’s retrospective first-passing action from one graph build to another. This measures response change without TCP or a learned predictor. Across four

**Terms at a glance. Tables and figures shorten SIFT-100K and Arxiv-Nomic-100K to SIFT and Arxiv.**

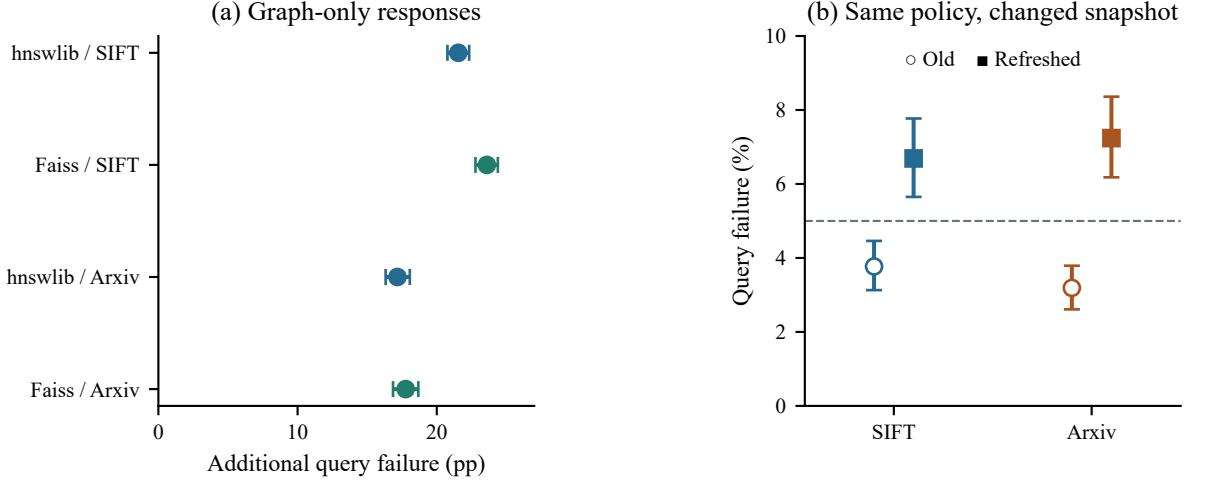
| Term             | Meaning                                                                    |
|------------------|----------------------------------------------------------------------------|
| ICBA             | Index-Conditioned Budget Audit: the measurement and decision workflow.     |
| TCP              | Target-calibrated pooling: the recurring-query history-pool recovery case. |
| NDC              | Number of distance calculations; not elapsed time.                         |
| CP / UCB         | Clopper–Pearson / upper confidence bound.                                  |
| Transport risk   | Target failure under a transferred action or policy and a declared event.  |
| LOTO             | Leave one target build out.                                                |
| Unresolved label | No registered action reaches the recall target.                            |
| Endpoint         | Largest grid action, whose risk is measured.                               |

hnsplib/Faiss–dataset combinations, target failure exceeds the target-specific reference by 17.17–23.60 percentage points. Second, after a 5% data refresh, we replay one executable history-pool policy: old-to-new risk rises from 3.77% to 6.69% on SIFT-100K and from 3.19% to 7.24% on Arxiv-Nomic-100K. Third, after freezing the analysis, we use previously untouched queries to test a simpler source-certified slack policy. It recovers useful work in Faiss but not across both hnsplib datasets. Figure 1 places the first two comparisons side by side; Section 7.3 reports the prospective boundary test.

*Index-Conditioned Budget Audit* (ICBA) connects these measurements to a serving decision. It identifies the permitted observations, fixes the failure event, isolates policy selection from certification, includes fallback in completed outcomes, and prices information acquisition. Its role is to prevent three distinct quantities from being interchanged: an observed recall improvement, a statistically qualified action, and a net economic benefit.

*Target-calibrated pooling* (TCP) is one recovery case within this audit. It takes historical budgets for the same recurring query, pools them conservatively, and chooses a global grid shift using target-selection queries. The case reveals a useful distinction: the original candidate/fallback procedure uses two individual 95% bounds, while a post-hoc allocation of one 5% confidence-error budget retains fewer candidates. Search savings remain positive under the latter, but Arxiv-Nomic’s mean reduction falls from 44.79% to 29.86%. The fresh-query test then asks a different question: can a policy fixed and certified using only source queries survive target builds? The answer depends on the implementation and its native action grid. Together, the tests expose the systems cost and the scope of certification.

We make three contributions. (1) A layered empirical analysis separates graph-only response non-portability, matched refresh-policy deterioration, and prospective source-policy transfer. (2) ICBA supplies an executable audit contract and distinguishes



**Figure 1: Two distinct portability tests. Left: graph-only transfer of retrospective first-passing actions; additional failure above each target’s own first-passing reference, with 95% query-cluster intervals conditional on 24 builds. Right: the same executable history-pool policy on matched old and 5%-refreshed builds; average query risk over ten builds, with crossed target-build/query 95% intervals. The old/new difference changes both graph and data. Neither comparison treats directed build pairs as independent trials.**

response evidence, fixed-target certification, completed fallback, and lifecycle value. (3) TCP quantifies target-calibrated recovery, while the fresh-query slack bridge gives a deliberately simpler boundary result: positive Faiss recovery and no cross-dataset hnswlib value. DARTH, Ada-ef, Vamana-style, and Deep1M analyses extend specific estimands without forming a common leaderboard.

The resulting systems lesson is to version the search policy with the index, query workload, and calibration record. Reusing an action is an operational choice, not a consequence of retaining its parameter name. Recovery can be substantial, but its useful form depends on the implementation, available observations, and serving horizon.

## 2 Related Work and Positioning

*Graph construction, evaluation, and maintenance.* HNSW combines hierarchical navigation with graph search, while DiskANN introduces Vamana as part of a system for large-scale search [12, 14]. Azizi et al. [4] compare graph-search methods and construction paradigms across datasets and scales. Elliott and Clark [8] show that insertion order and intrinsic dimensionality affect HNSW recall. These studies establish construction sensitivity. We examine a different experimental unit: an action or policy transferred from a source build to a target, with target difficulty separated from additional transfer loss. FreshDiskANN supports streaming updates [13]; IP-DiskANN performs insertions and deletions in place [16]; Wolverine repairs monotonic search paths disrupted by deletion [11]. These are maintenance alternatives that change the graph itself. ICBA instead evaluates the validity of a budget decision on the resulting graph. Our refresh experiment does not compare the maintenance efficiency of these alternatives.

*Adaptive work allocation and declarative recall.* LAET predicts query-specific termination conditions using both static and intermediate-search features [10]. DARTH instead predicts achieved recall during execution and adapts prediction intervals to terminate HNSW or IVF searches at a declarative target [5]. ANNiE estimates the work required to reach a recall target from graph-derived features; its ANNiE-S search uses those estimates directly [15]. Ada-ef models query-to-dataset similarity distributions to choose an HNSW exploration parameter [18]. It also studies incremental updates to its statistics and estimation table, including comparisons with stale and fully recomputed variants after insertions or deletions. Our question concerns the validity and cost of carrying a decision between builds. It applies to adaptive policies as well as fixed budgets, but comparisons must retain each policy’s permitted signals and update procedure. The frozen DARTH and Ada-ef bridges are evidence about those integrations, rather than a common ranking of all published methods.

*Quality guarantees and their target quantities.* ConANN applies conformal risk control to IVF search, controlling expected false-negative rate while adapting the number of probed clusters [9]. Its objective is the expected fraction of missed neighbors, whereas our principal event is whether a query falls below a recall threshold. ANNiE’s optional recall-target guarantee addresses the fraction of queries reaching the requested recall using quantile-loss minimization [15]. The DARTH+ preprint also introduces optional probabilistic quality guarantees for HNSW and IVF [6]. We study the build and policy covered by such an assessment, the information reused or reacquired, and the resulting behavior after rebuilding. The risk quantities remain explicit in comparisons.

*Calibration, selection, and the role of ICBA.* Conformal risk control bounds an expected monotone loss under its calibration assumptions [3]. Learn then Test formulates risk-controlled parameter selection as multiple hypothesis testing [2]. These are statistical foundations for auditing a search decision, rather than guarantees that automatically survive a changed execution environment. ConANN explicitly aligns calibration with index construction and assumes exchangeable query data [9]. In our setting, rebuilding can change the loss induced by the search even when the query population is retained. ICBA makes the target build, failure event, candidate selection, and fallback rule explicit. TCP supplies a concrete query-profile reuse and recovery case within that audit. This separation lets us ask whether target information restores useful search efficiency without conflating empirical recovery, a pipeline-level certificate, and net lifecycle savings.

### 3 Portability, Risk, and the Scope of Guarantees

#### 3.1 Execution Risk and Retrospective Responses

Let  $G$  specify a vector snapshot, graph, and search implementation, and let  $\mathcal{A} = (a_1 < \dots < a_j)$  be its native action grid. A policy  $\pi$  maps permitted query and history observations to an action. For  $q \sim \mathcal{Q}$ ,

$$\begin{aligned} Z_G(q, a) &= \mathbf{1}\{\text{Recall}@k(G, q, a) < \tau\}, \\ \mathcal{R}_G(\pi) &= \mathbb{E}_q Z_G(q, \pi(q)). \end{aligned} \quad (1)$$

Truth is computed against the corresponding snapshot. We distinguish target transport risk  $\mathcal{R}_{G_t}(\pi_s)$ , source risk  $\mathcal{R}_{G_s}(\pi_s)$ , and a reference-relative increment

$$\Delta_{s \rightarrow t} = \mathcal{R}_{G_t}(\pi_s) - \mathcal{R}_{G_t}(\pi_t^{\text{ref}}). \quad (2)$$

A matched old-to-new comparison instead subtracts  $\mathcal{R}_{G_s}(\pi_s)$ , keeping the policy fixed. These two differences answer different questions.

A retrospective first-passing label is  $B_G^{\text{first}}(q) = \min\{a_j : Z_G(q, a_j) = 0\}$ . A stable-tail label is

$$B_G^{\text{tail}}(q) = \min\{a_j : Z_G(q, a_\ell) = 0 \text{ for every } \ell \geq j\}. \quad (3)$$

Both require query truth; an empty set is  $+\infty$ , an unresolved label. Native action order need not make either raw recall or realized NDC monotone. Consequently, an action larger than a first-passing label need not succeed, whereas a finite stable-tail label establishes success above it on the recorded grid. The graph-only study uses first-passing labels and evaluates the transferred action's *actual* target response, rather than replacing execution failure by a budget-order comparison.

#### 3.2 Information Limits Explain What Must Be Observed

The following classical two-point argument [17] states why source information alone may be insufficient. For two environments  $E_0, E_1$ , let  $O$  include everything available to the decision rule, with laws  $P_0^O, P_1^O$ . Suppose acceptable risk-cost decisions form disjoint sets  $D_0, D_1$ , and loss satisfies  $L_i(a) \geq \lambda \mathbf{1}\{a \notin D_i\}$  for  $\lambda > 0$ . Then

every common decision rule satisfies

$$\max_i \mathbb{E}_i L_i(A) \geq \frac{\lambda}{2} (1 - \text{TV}(P_0^O, P_1^O)). \quad (4)$$

Indeed, deciding whether  $A \in D_0$  defines a test of the environment; its sum of error probabilities is at least  $1 - \text{TV}$ . If a common expensive endpoint is acceptable, disjointness fails and the bound can be vacuous. For random queries,  $O$  must include the query jointly with its transcript. Our measurements establish response differences, not an estimate of full-observation total variation or an impossibility result for every policy. The bound explains the information question; the audit below supplies the operational test.

#### 3.3 Certifying a Fixed-Target Decision

For a policy fixed independently of  $n$  i.i.d. target-certification queries,  $x$  failures yield the one-sided Clopper-Pearson bound [7]

$$U(x, n; \alpha) = \begin{cases} \text{Beta}^{-1}(1 - \alpha; x + 1, n - x), & x < n, \\ 1, & x = n. \end{cases} \quad (5)$$

Acceptance requires  $U \leq \delta$ . Confidence error  $\alpha$  and allowed query risk  $\delta$  are distinct. At zero failures,  $U = 1 - \alpha^{1/n}$ ;  $\alpha = \delta = 0.05$  requires at least 59 queries for acceptance to be possible. With 500 certification queries, nonzero failures can be accepted, but rejection can still reflect limited power near the boundary.

**PROPOSITION 3.1 (JOINTLY QUALIFIED SELECTION ON ONE TARGET).** *Freeze  $K$  policies before certification, and allocate  $\alpha_j > 0$  with  $\sum_{j=1}^K \alpha_j \leq \alpha$ . On a common i.i.d. certification sample, compute  $U_j = U(x_j, n; \alpha_j)$ . Select any policy with  $U_j \leq \delta$ , or abstain if none qualifies. Then*

$$\Pr\{a \text{ policy is selected and } \mathcal{R}_G(\pi_{\text{selected}}) > \delta\} \leq \alpha. \quad (6)$$

**PROOF.** By a union bound,  $\Pr\{\exists j : \mathcal{R}_G(\pi_j) > U_j\} \leq \sum_j \alpha_j$ . Otherwise every selectable policy satisfies the risk limit. Independence among policies' bounds is unnecessary.  $\square$

This is simultaneous risk control [2]. Selection may define the policies using a separate role; the proposition then applies conditionally on selection. A fallback must be included or separately justified. The statement is per fixed target, not simultaneous over a campaign, and does not bound risk conditional on the event of acceptance. The i.i.d. query interpretation concerns the represented workload; fixed-profile replay by itself proves no guarantee for a changed query population.

#### 3.4 Why Pooling Is Not the Same Certificate

For  $m$  source builds and one target with exchangeable budget labels at a fixed query, conformal rank reasoning [1] gives a different guarantee. Let  $r = \lceil (1 - \eta)(m + 1) \rceil$ . Use the  $r$ th source order statistic if  $r \leq m$  and it is finite; otherwise abstain. Then

$$\Pr\{a(q) < \infty, B_t(q) > a(q)\} \leq \frac{m + 1 - r}{m + 1} \leq \eta. \quad (7)$$

This is build-marginal label coverage for a fixed query. It translates to execution failure for stable-tail labels, but not automatically for first-passing labels. Nine sources offer rank resolution 1/10, and old/refreshed builds need not be exchangeable. Accordingly, TCP's

nine-history replay derives its target qualification from Equation (6), not from a 5% conformal claim. The short supplement supplies the rank proof and a margin-based selection result, and states which assumptions are uninstantiated in the experiment.

## 4 ICBA and Two Recovery Policies

### 4.1 The Audit Contract

ICBA leaves graph construction and native traversal unchanged. Its input contract names the source and target snapshots, implementation, query population, action grid, permitted observations, failure event, and reference policy. Its output records raw reuse, the selected action, certification counts and bounds, actual fallback, execution risk, and lifecycle cost. These fields also expose when two experimental settings cannot be quantitatively combined.

Target information has three separate uses. Selection chooses a policy using its assigned queries. Certification checks that frozen policy and any fallback; it does not retrain or reorder them. Evaluation measures the completed decision, including fallback executions. A raw-reuse row is retained even if it would fail certification, so a successful endpoint cannot hide an unsuccessful transferred policy. If no policy qualifies, the audit outputs abstention; diagnostic endpoint execution is recorded as unqualified, not approved deployment.

### 4.2 TCP: History Pooling Followed by One Target Shift

The recovery experiment uses native HNSW actions

$$\mathcal{A} = (10, 20, 40, 80, 120, 160, 200). \quad (8)$$

For each query and old build, encode the first-passing grid index as  $\tilde{j}_g(q)$ ; if none passes, store  $J$  while retaining the unresolved flag. For target seed  $t$ , pool the nine old builds other than the matched seed:

$$j_0^{(t)}(q) = \max_{g \in \mathcal{H}_t} \tilde{j}_g(q), \quad \pi_s(q) = a_{\min\{j_0^{(t)}(q)+s, J\}}. \quad (9)$$

The same-query old profiles exist for all three target roles. Thus TCP operates on a recurring, profiled workload, not unseen-query budget prediction. Storing a pooled index costs  $O(Q)$  for  $Q$  profiles; a serving decision is a lookup, addition, and clipping before native search. Acquiring those profiles is a separate, potentially dominant expense.

Target selection scans  $s = 0, \dots, J - 1$  in ascending order and takes the first shift whose selection-role CP bound at  $\alpha = 0.05$  is at most 0.05. If none passes, it returns the endpoint shift. This is a selection heuristic, not a multiple-testing certificate or a minimum-NDC optimizer. Both the selected policy and the endpoint are then fixed before certification.

Throughout this paper, TCP denotes the *target-calibrated pooling replay* in Equation (9). Its guarantee is determined by the target selection and certification procedure below, not by an exchangeable-build assumption.

### 4.3 Original Decisions and a Joint-Error Sensitivity

The original replay checks the selected candidate and endpoint separately at  $\alpha = 0.05$  each. It accepts the candidate if its bound passes, otherwise executes the endpoint with the endpoint’s qualification flag. The two allocations sum to 0.10, so they do not provide a 95% guarantee for the combined selection procedure.

We therefore add a *post-hoc frozen-response sensitivity*: keep every selected shift unchanged, use  $\alpha_c = \alpha_e = 0.025$ , and apply the same candidate-first order. If neither policy passes, abstain. This instantiates Proposition 3.1 at total  $\alpha = 0.05$  under its query-sampling assumptions, separately for each target. It does not rerun selection, enlarge the action class, or use evaluation to choose a new threshold. Its results are reported alongside the original decisions, not presented as a preregistered replication.

Figure 2 shows the data flow. The audit is reusable independently of TCP: a learned termination rule or a fixed budget can occupy the candidate input, provided its observation contract and failure event are retained.

### 4.4 A Source-Certified Slack Bridge

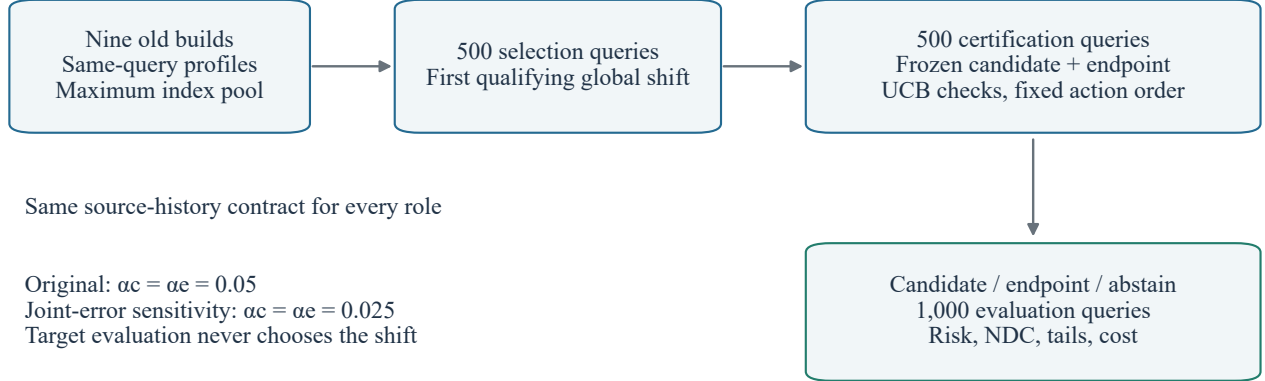
The second recovery policy removes TCP’s target-selection step. Before the fresh-query stage, a frozen source role of 375 queries per build scans the implementation’s six native actions in ascending order. It sets  $a_s$  to the lowest action whose one-sided Clopper-Pearson bound is at most  $\delta = 0.05$  under a Bonferroni allocation of  $0.05/6$ ; if no action passes, it uses the endpoint. This rule fixes one action for each of the 24 sources in each implementation–dataset block. The registered +0, +1, +2 lanes then shift  $a_s$  upward by the stated number of native grid levels and clip at the endpoint.

On 500 previously untouched source-certification queries, each frozen lane and its endpoint receive error allocations  $\alpha_c = \alpha_e = 0.025$ . A source decision executes the lane only when both bounds are at most  $\delta = 0.05$ ; otherwise it falls back to the certified endpoint. Another 500 untouched queries evaluate the completed source-to-target decision over all non-diagonal build pairs. The source actions, role identifiers, lane definitions, and their preregistration digest were fixed before any target-evaluation vector or truth was accessed.

This bridge uses no target labels for design, certification, or action selection, and it does not use the retrospective stable-tail action or evaluation feedback. Target truth is accessed only for the held-out audit. The three lanes are separate preregistered estimands rather than a best-after-observation search. Consequently, the experiment can show that a particular fixed slack transfers within a registered implementation–grid family, but not that the lane is uniformly optimal or jointly selected at 95% confidence across implementations.

## 5 Experimental Design and Inference

*Graph-only rebuilds.* Each dataset contains 100,000 base vectors. SIFT-100K uses dimension 128 and Euclidean distance; Arxiv-Nomic-100K uses dimension 768 and normalized inner-product search. Each hnsplib dataset has 24 builds crossing eight seeds and three insertion orders (random, ascending local intrinsic dimensionality, and descending). The clean Faiss replay has 24



**Figure 2: ICBA applied to TCP. Same-query old-build profiles define the pooled policy for each target role. Selection chooses one global shift; certification checks the frozen candidate and endpoint; evaluation has no feedback path. The original replay assigns 0.05 to each check. The joint-error sensitivity assigns 0.025 to each, allowing a per-target union bound of 0.05.**

registered builds per dataset. Each uses 750 evaluation queries, all 552 non-diagonal directed build pairs, and unchanged base membership and truth. The hnsplib grid is (10, 20, 40, 80, 120, 200); the Faiss grid is (16, 32, 64, 128, 256, 512). Their native actions are not equated numerically.

For this study, the source response rule chooses  $B_s^{\text{first}}(q)$ , or the maximum action if unresolved. The target reference chooses  $B_t^{\text{first}}(q)$  when finite. Its failure is the target-unresolved event; transport failure is either that event or failure of the actual target response at the source-derived action. This convention preserves unresolved mass rather than dropping difficult queries. It is a retrospective response experiment, not a deployable oracle algorithm or an independently source-certified policy.

*Refresh and executable-policy replay.* A separate seed-991 membership change replaces 5% of each 100K collection with previously excluded vectors, preserving cardinality. Truth is refreshed. Ten old and ten refreshed builds per dataset are replayed through the official DARTH Faiss-HNSW fork, using graph degree parameter  $M = 16$ , construction  $ef = 100$ , one search thread,  $k = 10$ , and Equation (8). For each matched pair, TCP excludes that seed from its nine-history pool. The old-to-new diagnostic executes the *same* unshifted policy on both snapshots. Old-side certification checks the pre-existing pooled policy; no source outcome is used to select a different pool.

Target roles contain 500 selection, 500 certification, and 1,000 evaluation queries, with disjoint content and the same IDs reused across builds. Checks against base vectors and insertion reserves accompany the role manifest. Source-history responses for each role precede target selection. Reusing historical profiles for evaluation IDs is part of the recurring-workload contract, not a learned predictor’s held-out generalization test. Seeds for graph construction, membership refresh, and statistical resampling are separate roles even where the integer 991 appears in two configurations.

*Fresh-query source-policy confirmation.* After the preceding analyses were frozen, 1,000 previously untouched query IDs per dataset were split in registered order into 500 source-certification and 500 target-evaluation queries, with zero overlap with the earlier 375-query source role or any prior evaluation role. We reuse the same 24 hnsplib and 24 Faiss indexes per dataset; no graph is rebuilt or mutated. The source rule, its 96 actions, the +0, +1, +2 lanes, and the query-role identifiers were sealed before accessing any of the 1,000 future-query vectors or truth. Candidate and endpoint receive  $\alpha = 0.025$  each, and target evaluation cannot change the action or fallback. Primary intervals resample the 500 target-query IDs while retaining each query’s complete 552 directed-pair vector. This is fresh-query confirmation conditional on the registered builds, not independent software replication or inference over an open build population.

*Metrics.* Risk is query failure under  $k = 10$ ,  $\tau = 0.95$ . Relative serving gain is

$$g = 1 - \frac{\bar{c}_{\text{policy}}}{\bar{c}_{\text{endpoint}}}, \quad (10)$$

a ratio of pooled means, equivalently a relative total-NDC reduction for equal query counts. We report query-pooled p95/p99 NDC and per-target tails, retaining every fallback execution. These are search-work measures, not latency measurements. Observed tail non-increase is distinguished from a formal noninferiority test.

*Inference and its sampling units.* Graph-only intervals resample 750 query IDs, retaining each query’s complete directed-pair vector, with builds fixed. They do not treat 552 overlapping directions as independent builds. For the refresh replay we recompute 5,000 crossed bootstrap replicates with seed 991: sample target-build indices, then sample one shared query-index vector used in *all* sampled builds. The paired policy/reference outcomes remain together. Percentile intervals reflect this product resampling, conditional on

the stored source histories and selected shifts; they are not confidence guarantees for a new build-generation population. Build-only and shared-query-only sensitivities appear in the supplement.

LOTO deletes a target without refitting, and deletion of the largest relative-gain target is one specified member of that sensitivity family. Acquisition-cost intervals resample target builds conditional on the finite profiling/query records. Vamana and external-method intervals retain their original units, identified beside the corresponding estimates. No interval is pooled across these protocols.

*Evidence order.* We answer nine questions in reading order: graph-only transfer (RQ1), matched refresh transfer (RQ2), audit decisions (RQ3), target-calibrated and source-only recovery (RQ4), tails (RQ5), build sensitivity (RQ6), external/family/scale scope (RQ7), economics (RQ8), and reproducibility (RQ9). The old responses and original decisions remain unchanged. Crossed inference, matched-source qualification, joint-error replay, and completed acquisition accounting are derived analyses of existing records; the fresh-query bridge is the separately frozen confirmation stage.

## 6 Where Budget Portability Fails

### 6.1 RQ1: Do Responses Transfer When Only the Graph Changes?

Retrospective first-passing actions frequently fail after a graph-only rebuild, even though collection membership, queries, and truth are unchanged. Table 1 reports absolute target risk, the unresolved-reference risk, and their difference. Across hnsplib and Faiss on both datasets, the increment ranges from 17.17 to 23.60 percentage points; every query-cluster interval is above zero. This is a response-level result independent of TCP’s pooling or recalibration.

The variation is principally among finite budgets: 75.87–91.07% of queries have different finite first-passing actions across builds, while only 0.53–2.67% switch between resolved and unresolved states. Removing the 1% of queries with largest incremental contribution leaves increments of 21.37/16.95 percentage points for hnsplib and 23.43/17.53 for Faiss, on SIFT/Arxiv respectively. Endpoint censoring and a handful of extreme queries therefore do not explain the result.

This experiment diagnoses instability of the budget–response relation, not universal failure of all source policies. Its source rule consults source-query truth, and the target reference consults target-query truth. Neither is offered as a deployed algorithm. Nevertheless, it establishes why carrying an action from one graph to another requires evidence beyond the fact that the action succeeded on its source graph. The mechanism established here is behavioral variation; attributing it to particular edges or navigation paths would require a separate structural analysis.

### 6.2 RQ2: Does an Executable Policy Deteriorate after Refresh?

The matched old/new replay holds the nine-history policy fixed and applies it to both snapshots. Unlike RQ1, this changes 5% of the vectors as well as rebuilding. SIFT’s empirical risk rises from

3.77% to 6.69%; Arxiv-Nomic’s rises from 3.19% to 7.24%. The corresponding paired increments [crossed 95% interval] are 2.92 [1.75, 4.07] and 4.05 [2.86, 5.31] percentage points. Thus this is a measured deterioration, not merely an absolute target risk with an unknown source baseline.

Independent single-policy CP checks at  $\alpha = 0.05$  qualify six of ten old SIFT policies and five of ten old Arxiv policies. All eleven fail the corresponding target check. These are per-policy checks and a post-hoc paired diagnostic; they are not a simultaneous certificate over eleven transitions. Target evaluation risk exceeds 5% in all ten builds of each dataset. Removing the highest-risk target leaves 6.61% and 7.13%, respectively. The observed refresh failure is distributed across targets rather than concentrated in one anomalous build.

The same policy information is held fixed in each old/new comparison, but the experiment does not separate changed graph paths from changed nearest-neighbor membership. Together, RQ1 and RQ2 establish complementary facts: unchanged data do not make the response function portable, and a recurring-query policy can lose its old qualification after refresh plus rebuilding.

## 7 Auditing and Recovering Rebuilt Workloads

### 7.1 RQ3: What Does the Audit Actually Authorize?

Auditing unshifted source reuse rejects every candidate and routes all 20 target decisions to the endpoint in the original replay. The audit therefore prevents cheap but unqualified reuse, yet recovers no search saving by itself. Target-selection recalibration changes the candidate before certification. The original procedure accepts 19 of 20 recalibrated candidates; Arxiv seed 1229 has candidate UCB 0.05056 and falls back despite being close to the threshold.

The joint-error sensitivity is more selective (Figure 3). With  $\alpha_c = \alpha_e = 0.025$ , it accepts all ten SIFT candidates and six Arxiv candidates. All twenty endpoints pass their allocated checks, so four Arxiv decisions fall back and none abstains. The final decision satisfies the per-target construction in Proposition 3.1, subject to its query-sampling conditions. Certification rejection is not a finding of true unsafety: it changes the supported action given the available 500-query certificate.

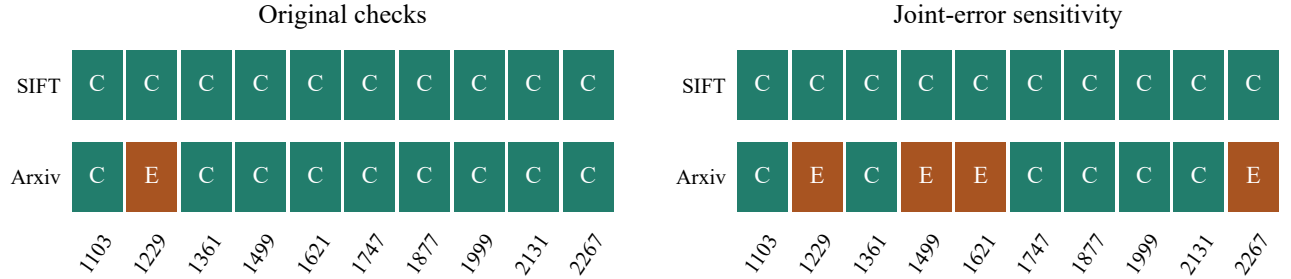
### 7.2 RQ4: How Much Search Efficiency Is Recovered?

Recalibration restores substantial search efficiency within the recurring-profile workload (Table 2). The original decisions reduce risk to 1.79% and 2.04%, with relative mean-NDC reductions of 44.39% and 44.79%. Crossed resampling widens SIFT’s interval compared with the old independently-within-build procedure: it is now [42.76, 46.01]%, rather than [43.88, 44.91]%. Preserving shared query dependence changes uncertainty even when the point estimate is unchanged.

Under the joint-error allocation, SIFT is unchanged, while Arxiv risk becomes 1.68% and relative mean-NDC reduction becomes 29.86 [14.75, 45.00]%. Figure 4 shows that stricter qualification moves the decision toward the endpoint: it reduces risk and sacrifices some efficiency. The positive intervals support recovery of

**Table 1: Graph-only response transfer at  $k = 10$ ,  $\tau = 0.95$ . Each row has 24 builds, 552 directed pairs, and 750 shared query IDs. Risks and finite-action variation are percentages; increments are percentage points [95% query-cluster interval], conditional on the fixed build set. The reference uses the target’s own first-passing response; its failures are unresolved queries. Finite variation counts queries with at least two distinct finite first-passing actions, divided by all 750 queries.**

| Implementation | Dataset | Absolute risk | Reference risk | Increment [95% interval] | Finite variation |
|----------------|---------|---------------|----------------|--------------------------|------------------|
| hnswlib HNSW   | SIFT    | 22.31         | 0.77           | 21.55 [20.76, 22.33]     | 89.33            |
| Faiss HNSW     | SIFT    | 23.67         | 0.07           | 23.60 [22.78, 24.39]     | 91.07            |
| hnswlib HNSW   | Arxiv   | 18.24         | 1.07           | 17.17 [16.33, 18.06]     | 75.87            |
| Faiss HNSW     | Arxiv   | 17.93         | 0.17           | 17.77 [16.86, 18.68]     | 75.87            |



**Figure 3: One cell per target decision, ordered by registered seed rather than a time axis. C: candidate accepted; E: endpoint fallback. Left: original two individual 95% checks. Right: post-hoc 0.025+0.025 joint-error allocation. The same selected shifts and certification queries are used in both. Candidate acceptance falls from 19/20 to 16/20, while every completed decision remains qualified by its corresponding endpoint or candidate check.**

**Table 2: Completed refresh decisions on ten targets and 1,000 evaluation queries per target. Query risk and relative mean-NDC reduction are percentages [crossed 95% interval]; intervals are conditional on fixed source histories and selected shifts. Original and joint-error rows differ only in certification composition. All fallback executions are included. Mean and p95/p99 are distance calculations, not latency.**

| Dataset | Decision               | Risk [95% interval] | Reduction [95% interval] | Mean NDC | p95   | p99   |
|---------|------------------------|---------------------|--------------------------|----------|-------|-------|
| SIFT    | Endpoint               | 0.97 [0.53, 1.50]   | 0.00 [0.00, 0.00]        | 1,826    | 2,615 | 2,855 |
| SIFT    | Direct reuse           | 6.69 [5.65, 7.77]   | 59.51 [57.83, 61.20]     | 739      | 2,132 | 2,706 |
| SIFT    | Original recalibration | 1.79 [1.25, 2.40]   | 44.39 [42.76, 46.01]     | 1,015    | 2,349 | 2,774 |
| SIFT    | Joint-error replay     | 1.79 [1.25, 2.40]   | 44.39 [42.76, 46.01]     | 1,015    | 2,349 | 2,774 |
| Arxiv   | Endpoint               | 1.01 [0.61, 1.47]   | 0.00 [0.00, 0.00]        | 2,397    | 3,252 | 3,504 |
| Arxiv   | Direct reuse           | 7.24 [6.18, 8.36]   | 64.21 [62.51, 65.85]     | 858      | 2,586 | 3,322 |
| Arxiv   | Original recalibration | 2.04 [1.45, 2.69]   | 44.79 [34.32, 50.96]     | 1,323    | 2,974 | 3,398 |
| Arxiv   | Joint-error replay     | 1.68 [1.10, 2.35]   | 29.86 [14.75, 45.00]     | 1,681    | 3,117 | 3,460 |

useful search work on both datasets, not a claim of optimality over adaptive-search methods with different information budgets. In particular, the comparator here is the same graph’s fixed endpoint, not the best target-trained method.

### 7.3 RQ4 Boundary Test: Can Source-Only Slack Transfer?

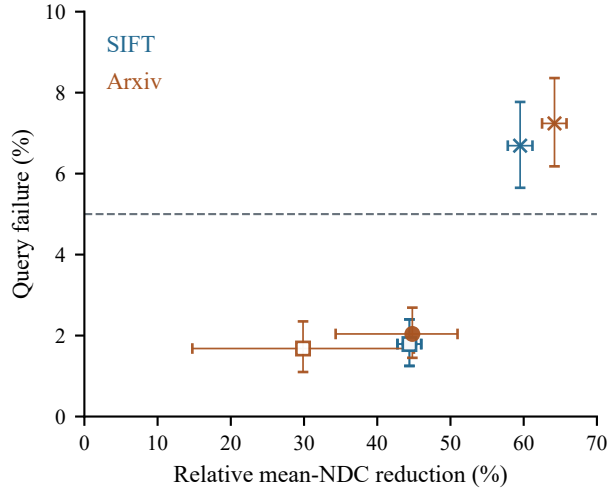
The fresh-query experiment tests the source-certified policy of Section 4.4, not TCP. Table 3 reports all three registered lanes.

Each row contains 24 independently certified source decisions and 552 held-out source–target evaluations. These are different objects: source certification determines the executable candidate or endpoint fallback, whereas target truth supplies an empirical portability audit rather than a target deployment certificate.

The Faiss +1 lane has empirical target risk 0.183 [0.083, 0.316]% on SIFT and 0.674 [0.280, 1.211]% on Arxiv-Nomic, with relative mean-NDC reductions of 8.27 [8.27, 8.28]% and 29.09 [29.06, 29.13]%. Its raw per-direction risks have median/maximum values of 0.0/1.4% and 0.6/1.6%, respectively, so the pooled estimates do

**Table 3: Fresh-query source-slack results, conditional on 24 registered builds per implementation–dataset block. Risk and relative mean-NDC reduction are percentages [95% query-bootstrap interval]. Source decisions show candidate executions (C) and certified-endpoint fallbacks (F). Target audit status reports directions whose held-out CP upper bound is below 5% (B), unresolved at 5% (U), or above 5% (A); these statuses are not target deployment certificates.**

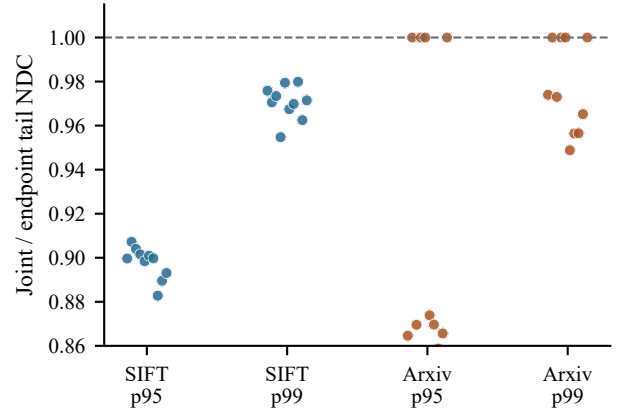
| Impl.   | Dataset | Lane | Source C/F | Target B/U/A | Empirical risk [95% CI] | NDC reduction [95% CI] |
|---------|---------|------|------------|--------------|-------------------------|------------------------|
| hnswlib | SIFT    | +0   | 24/0       | 543/9/0      | 1.49 [0.84, 2.26]       | 1.31 [1.31, 1.32]      |
|         |         | +1   | 24/0       | 552/0/0      | 1.40 [0.76, 2.16]       | 0.00 [0.00, 0.00]      |
|         |         | +2   | 24/0       | 552/0/0      | 1.40 [0.76, 2.16]       | 0.00 [0.00, 0.00]      |
| hnswlib | Arxiv   | +0   | 24/0       | 552/0/0      | 1.32 [0.60, 2.16]       | 0.00 [0.00, 0.00]      |
|         |         | +1   | 24/0       | 552/0/0      | 1.32 [0.60, 2.16]       | 0.00 [0.00, 0.00]      |
|         |         | +2   | 24/0       | 552/0/0      | 1.32 [0.60, 2.16]       | 0.00 [0.00, 0.00]      |
| Faiss   | SIFT    | +0   | 24/0       | 478/74/0     | 1.30 [0.85, 1.86]       | 53.81 [53.76, 53.85]   |
|         |         | +1   | 24/0       | 552/0/0      | 0.18 [0.08, 0.32]       | 8.27 [8.27, 8.28]      |
|         |         | +2   | 24/0       | 552/0/0      | 0.06 [0.01, 0.13]       | 0.00 [0.00, 0.00]      |
| Faiss   | Arxiv   | +0   | 22/2       | 528/24/0     | 1.72 [1.10, 2.47]       | 58.21 [58.17, 58.26]   |
|         |         | +1   | 24/0       | 552/0/0      | 0.67 [0.28, 1.21]       | 29.09 [29.06, 29.13]   |
|         |         | +2   | 24/0       | 552/0/0      | 0.36 [0.08, 0.83]       | 0.00 [0.00, 0.00]      |



**Figure 4: Refresh risk–work tradeoff with crossed 95% intervals. The 5% line is the query-risk target, not a confidence bound. Filled symbols: original recalibration; open symbols: joint-error replay; crosses: direct reuse. SIFT’s original and joint points coincide.**

not conceal a direction above 5%. The lower saving remains positive after leaving out any target build, and p95/p99 stay below the endpoint. The narrow gain intervals resample 500 query vectors with their complete 552-direction outcomes; they do not count directions as independent observations.

The native grids explain the implementation boundary. In hnswlib, 23 of 24 SIFT sources and all 24 Arxiv sources already select the endpoint; +1 clips the remaining SIFT source, so both +1 and +2 equal endpoint execution. In Faiss, +1 clips 20 of 24 SIFT sources and 10 of 24 Arxiv sources, leaving non-endpoint actions that produce the observed savings; +2 reaches the endpoint everywhere. The larger +0 savings come with 74 unresolved SIFT directions and 24 unresolved Arxiv directions in the held-out audit. Since one native rung represents different work increments in the



**Figure 5: Per-target tail ratios for joint-error replay, relative to that target’s endpoint; ten dots per dataset/quantile. Equality reflects an endpoint fallback. All ratios are at most one. These descriptive tail comparisons are not formal non-inferiority certificates or wall-clock results.**

two libraries, this is an implementation-and-grid result, not a claim that one HNSW implementation is intrinsically more portable.

#### 7.4 RQ5: Does Recovery Worsen the Search-Work Tail?

Neither recalibration rule increases query-pooled p95 or p99 relative to the endpoint. For joint-error replay, p95 is 2,349 versus 2,615 NDC on SIFT, and 3,117 versus 3,252 on Arxiv; p99 is 2,774 versus 2,855 and 3,460 versus 3,504. Every individual target also has non-increasing p95 and p99. Figure 5 displays the per-target ratios without connecting unrelated seeds.

The fallback is a build-level choice, so its serving distribution is exactly the fixed endpoint’s distribution for that build. Combining it with candidate-serving builds can change pooled quantiles even without changing a candidate’s query path. Reporting both pooled and per-target tails avoids hiding this composition effect. It

also avoids inferring tail behavior from the much larger mean gain alone.

### 7.5 RQ6: Is the Benefit Driven by One Target?

The minimum LOTO mean-NDC reduction is 44.37%/44.23% for original SIFT/Arxiv decisions, and 44.37%/27.64% for joint-error decisions. Deleting the largest relative-gain build produces those same minima in these records. These are overlapping checks, not independent confirmations. Target-only and shared-query-only interval sensitivities in the supplement show different uncertainty sources: SIFT’s variation is dominated by queries, whereas Arxiv’s stricter-rule result also varies strongly with the candidate/fallback choice across targets. The retained positive direction supports a fixed-target-set recovery finding; it does not certify an unseen rebuild distribution.

## 8 RQ7: Scope across Methods, Scale, and Graph Family

*External adaptive methods.* The frozen DARTH adapters produce target query-failure rates of 22.55% on SIFT-100K and 23.12% on Arxiv-Nomic-100K at Recall@10 < 0.95. Ada-ef yields 9.52 [9.44, 9.60]% on ten Arxiv targets, with a target-build bootstrap conditional on its query set. All ten Ada-ef candidate certificates fail; audited execution uses the qualified endpoint and has zero search gain relative to that endpoint. Ada-ef is absent from Euclidean SIFT because the inspected estimator implements cosine/inner-product semantics, not a semantics-preserving L2 path. Its exclusion is an implementation-scope boundary, not a failed SIFT result.

These measurements show why query-risk qualification matters beyond TCP. They do not, without matched source checks, identify how much failure was caused by rebuilding, and the event is not interchangeable with mean recall or expected false-negative fraction. They therefore remain qualification studies rather than a ranking of published methods. The TCP comparator and information budget are stated separately.

*Larger scale.* On Deep1M, eight hnsplib builds cross four seeds with two insertion orders, giving 56 directed pairs on 1,000 queries. Source-derived first-passing actions have 28.86% transport risk against 8.82% target-reference risk. The incremental risk is 20.04 [17.29, 22.77] percentage points, using a target-build bootstrap conditional on the frozen source histories and query set. Keeping the reference risk visible matters: this is additional loss above a difficult, partly unresolved target reference, not a transfer from a zero-risk baseline. It extends response non-portability to the one-million-vector scale.

*Vamana-style evidence.* Six source and six separate target builds produce 36 directions and 750 queries per dataset, using native search-list size and beam width one. The frozen event table combines its stage-specific under-budget and censoring outcomes. A target-build reanalysis gives 16.86 [15.88, 17.93]% on SIFT and 11.37 [10.80, 11.99]% on Arxiv. Its recorded responses do not reconstruct exactly the first-passing execution estimand used for hnsplib/Faiss. We retain it as directional cross-family evidence, not another row of Table 1.

Vamana cost-tax intervals include zero: 1.07 [−1.03, 2.86]% and 1.17 [−0.83, 3.03]%, respectively, for the mean of per-target ratios in that reanalysis. Thus it supplies no established positive cost penalty or cross-engine superiority. Table 4 summarizes which question each setting can answer.

Together with RQ1, these results support a layered conclusion: response non-portability is seen across HNSW implementations and at larger scale; qualification of adaptive actions is operationally consequential; and a distinct graph family supplies supporting evidence with narrower semantic comparability. They do not establish uniform failure across every graph family or every adaptive policy.

## 9 RQ8: When Does Recovery Repay Its Information Cost?

Search savings repay recovery only over a sufficiently long recurring workload. We express the accounting in native distance evaluations,

$$C_{\pi}(N) = C_{\text{acquire},\pi} + N\bar{c}_{\pi}, \quad N_{\text{BE}} = \frac{\Delta C_{\text{acquire}}}{\bar{c}_{\text{end}} - \bar{c}_{\pi}}. \quad (11)$$

If incremental acquisition is positive but serving saving is not, no finite break-even exists. Graph rebuilding is common to both policies on the same target. Control adds no distance calls but is not assigned zero elapsed time.

*Two explicit acquisition scenarios.* With *cached history*, old query profiles and their truth are sunk costs. We charge the complete target-selection grid, the selected candidate’s certification searches, additional endpoint-certification searches where the actions differ, and exact target truth for the 1,000 selection/certification queries. Truth is modeled as one scan of 100,000 vectors per query. With *cold complete history*, we also charge sequential old-build profiling for all three roles on all nine contributing histories and exact old-snapshot truth for all 2,000 query IDs. That truth is reusable across old builds of the same collection and is counted once, not nine times. These are workload-acquisition scenarios, not newly measured elapsed-time costs.

We conservatively charge all endpoint certification to recovery, although the reference may also need some of this work. We price the evaluated decision rule without charging repeated calls for the same action and query twice. The earlier partial calculation included old profiling only for evaluation IDs and omitted old truth and extra endpoint certification. Completing these items materially changes the economic conclusion.

For joint-error replay, cold-history break-even is approximately 396,000 executions on SIFT and 457,000 on Arxiv, versus 129,000 and 148,000 with cached history (Table 5). Four Arxiv targets fall back and never repay a positive recovery overhead through serving savings. The original procedure has lower cold-history break-even on Arxiv, approximately 305,000, but different certification composition. It is therefore insufficient to attach the original economics to the stricter decision rule.

At  $N = 10^6$ , the cold-history joint-error model yields mean net savings of about 490 million NDC on SIFT and 389 million on Arxiv. Their target-build intervals are [488, 491] million and [31, 747] million, respectively. At  $10^5$ , both cold-history means are negative.

**Table 4: Scope of the extension evidence. Each row retains its event, reference, and inference unit. All numeric intervals in the text are 95%; no common cross-engine ranking or pooled effect is formed.**

| Setting      | Event and reference                                                                      | Supported comparison                                                        | Unit / query count                    |
|--------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------|
| DARTH        | Absolute target Recall @10 < 0.95 risk; no matched reference subtraction                 | Target qualification under the frozen adapter                               | Target summaries; 1,000/build         |
| Ada-ef       | Absolute target Recall @10 < 0.95 risk; endpoint for audited saving                      | Candidate rejection versus endpoint execution on Arxiv                      | 10 target builds; 1,000/build         |
| Deep1M       | Source-derived execution versus target first-passing reference, unresolved mass retained | Additional response-transfer loss at 1M scale                               | 8 target builds; 1,000 shared queries |
| Vamana-style | Frozen stage under-budget/censoring event; stage-specific cost reference                 | Directional cross-family transfer evidence, not a harmonized HNSW increment | 6 target builds; 750 shared queries   |

**Table 5: Joint-error recovery: break-even query executions in thousands [95% target-build interval], conditional on the finite acquisition workload. The ratio of mean overhead to mean saving is not a per-build guarantee. “No saving” counts nonamortizing targets.**

| Dataset | History | Break-even, $10^3$ | No saving |
|---------|---------|--------------------|-----------|
| SIFT    | Cached  | 129 [129, 129]     | 0/10      |
| SIFT    | Cold    | 396 [395, 397]     | 0/10      |
| Arxiv   | Cached  | 148 [99, 296]      | 4/10      |
| Arxiv   | Cold    | 457 [305, 915]     | 4/10      |

**Figure 6: Joint-error recovery, modeled net distance savings per target versus serving horizon. Solid: cold complete history; dashed: cached history. Dots mark reported horizons; lines depict the stated affine cost model, not additional measurements. Endpoint-fallback targets remain in the average.**

Figure 6 shows how the crossover depends on what information already exists.

This calculation assumes repeated executions over the profiled workload. A cache miss requires new information or a declared

endpoint action; an unbounded stream of new query identities can invalidate the amortization. Acquisition of historical graphs themselves is not reconstructed here: cold history means acquiring profiles on retained graphs, not rebuilding the entire archive. Memory traffic, storage, energy, and control time require separate prices. The result is a conditional distance-work benefit, not an end-to-end latency or monetary claim.

## 10 Reproducibility and Operational Implications

### 10.1 RQ9: Can the Decisions and Numbers Be Checked?

The compact evidence package contains per-target certification failures, both confidence allocations, selected shifts, paired evaluation arrays, cost components, the fresh-query summary, and generated table inputs. One script regenerates figures and typeset rows; another reanalyzes frozen native-response files through explicit repository and replay-root arguments. It checks the original twenty target decisions before deriving the new sensitivities. Historical records and the prospective source-slack stage remain separately labeled.

The preceding artifact audit preserved 80 pre-audit scientific files, passed a neutral-path smoke and 28 focused tests, and exposed guarded replay entry points for DARTH, Vamana, and Deep1M. In a clean Python environment built from the anonymous allowlist alone, the manuscript package passes 244 evidence checks, reproduces all generated table text exactly, regenerates six figures, and verifies 210/210 fresh-query checks. It also includes the 96 frozen source actions, a sanitized preregistration summary, and the source-slack protocol. These are reference-environment checks, not independent native-search replication. Recreating native searches still requires separately provisioned vectors, truth, indexes, binaries, and role manifests. The short supplement gives the statistical recipe and detailed certificate tables; the package maps each claim to its compact evidence source.

*Interpretation for index maintenance.* The experiments suggest three operator decisions. After a graph-only rebuild, a historical budget is evidence about a source response, not the new one. After refresh, test the unchanged policy on the target rather than interpreting high mean recall as query-risk qualification. If recovery is considered, evaluate its completed candidate/fallback procedure and the horizon needed to pay for information. ICBA makes these

checks explicit; TCP demonstrates target-calibrated recovery on a recurring workload, while the fresh-query bridge shows when a cheaper source-only rule is or is not sufficient.

*Remaining boundaries.* The graph-only result is a retrospective response diagnostic; TCP’s positive evidence is a Faiss-HNSW recurring-profile case on two 100K datasets. The fresh-query bridge is prospective with respect to query roles but conditional on existing registered builds, and its positive economic result is specific to the registered Faiss action grid. Deep1M extends response evidence, not recovery economics. The certification sensitivity is post-hoc and per target. The experiments do not identify a causal navigation-edge mechanism, demonstrate a general unseen-query predictor, or measure controlled wall-clock recovery. Dynamic graph repair and target-trained adaptive methods are alternatives, not defeated baselines. These boundaries identify where follow-up work would add evidence rather than repeat the audit.

## 11 Conclusion

Search budgets are part of an index’s operational state. On unchanged collections, source-derived first-passing actions incur substantial additional failure across hnsplib and Faiss rebuilds. After refresh, an unchanged recurring-query policy deteriorates and target recalibration recovers useful work. On fresh queries, a simpler source-certified slack rule transfers economically on the registered Faiss grids but not on the hnsplib grids. ICBA makes these outcomes comparable without conflating them: stricter qualification can increase fallback, and complete acquisition accounting can delay break-even. Portability, certification, and economic value must therefore be decided for the target index, implementation, action grid, and workload.

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